

# EXECUTIVE WORKFORCE ANALYSIS REPORT

## Predictive Risk Modeling: Proactive Identification of Workforce Attrition Patterns

### 1. Executive Summary

To replace reactive exit tracking with a proactive retention strategy, we engineered an AI-driven predictive model analyzing historical patterns across 1,470 corporate profiles. The system successfully maps underlying organizational stress points to catch flight risks before resignation papers are filed.

### 2. Core System Performance

The predictive model was optimized to catch risks early, capturing **65.96% of all true future employee departures**. While this high sensitivity generates occasional safe false alarms (flagging loyal employees out of caution), it narrows the target risk pool down effectively, giving HR a vital intervention window to deploy retention strategies.

### 3. Primary Workforce Vulnerabilities Discovered

- **The Overtime Burnout Multiplier:** Consistent overtime emerged as the strongest predictor of resignation. Staff working extra hours leave at an accelerated rate regardless of their salary tier, proving that workload fatigue consistently overrides baseline compensation.
- **Critical Departmental & Role Exposure:** The Sales department faces the highest organizational attrition at **20.63%**. At a granular level, entry-level **Sales Representatives** exhibit a severe exit rate of **39.76%** (nearly 2 out of 5 leave), followed by Laboratory Technicians at **23.94%**.
- **The Compensation Gap:** Employees who resigned earn an average monthly income of **4,787.09**, which is **29.93% lower** than the average compensation profile of retained staff (**6,832.74**). Turnover is heavily concentrated in entry-to-mid wage tiers.

### 4. Actionable Strategic Recommendations

- **Enforce an Operational Overtime Cap:** HR must collaborate with leadership to enforce hard ceilings on continuous overtime hours. Additionally, department managers' performance evaluations should be directly tied to the retention health of their immediate teams.
- **The 3-Year Tenure Stay-Interview Mandate:** The data highlights a major workforce flight window occurring precisely during an employee's **third year** with the enterprise. HR should bypass traditional annual review cycles and deploy targeted "Stay Interviews" the moment mid-tier staff record dipping satisfaction metrics.

### 5. Model Limitations

This model functions strictly as an internal diagnostic radar based on historical trends. It cannot account for unrecorded external variables such as aggressive competitor poaching campaigns, macroeconomic shifts, or sudden personal lifestyle transitions. It serves to guide, not replace, strategic human leadership.